

AZURE DATA ENGINEERING JOURNEY

SQL Fundamentals

DDL · DML · CRUD — hands-on practice notes
from today's session.

SWIPE TO LEARN →

DDL · DATA DEFINITION LANGUAGE



CREATE

01

CREATE DATABASE sets up a brand-new workspace, and **CREATE TABLE** defines its structure — column names and data types. Every SQL project begins right here.

```
CREATE DATABASE AZURE_DB
USE AZURE_DB
CREATE TABLE EMPLOYEES (
    EMP_ID INT,
    EMP_NAME VARCHAR(30)
)
```

DML · DATA MANIPULATION LANGUAGE

02

⊕ INSERT

INSERT INTO adds new rows to a table. Multiple records can be added in a single statement — fast, clean bulk data entry.

```
INSERT INTO EMPLOYEES (EMP_ID, EMP_NAME)
VALUES
  (1, 'RAHUL'),
  (2, 'AJAY'),
  (3, 'AJITH')
```

DDL + DML · STRUCTURE VS DATA

ALTER vs UPDATE

03

Two very different kinds of change — one edits the table's blueprint, the other edits what's inside it.



ALTER TABLE

Changes the table's **structure** — add new columns like SALARY or CITY.

```
ALTER TABLE EMPLOYEES  
ADD SALARY INT
```



UPDATE

Changes the **data** inside existing rows using SET and WHERE.

```
UPDATE EMPLOYEES  
SET SALARY = 85000  
WHERE EMP_NAME = 'AJITH'
```

REMOVING DATA & OBJECTS

DELETE vs TRUNCATE vs DROP

Three commands, three levels of removal — know exactly which one you need.



DELETE

Removes specific rows with **WHERE**.
Can be rolled back.
Table structure stays intact.



TRUNCATE

Removes **all rows** at once — faster than DELETE. Resets identity, keeps the structure.



DROP

Deletes the **entire table** — structure and data — permanently from the database.

QUICK RECAP

SQL = CRUD

CREATE → INSERT

READ → SELECT

UPDATE → UPDATE

DELETE → DELETE · TRUNCATE · DROP

Following my Azure Data Engineering journey — one query at a time.

💬 Let's connect, and feel free to repost if this helped your revision too!